Week 5 - Atmospheric Dynamics: Worked Answers

EART23001 Atmospheric Physics and Weather

Workload guide. The shaded question headings show the intended workload:

- **IN CLASS**: selected anchor problems that we may work through together in the timetabled session.
- **YOU SHOULD TRY**: expected independent practice after the session.
- **EXTRA CHALLENGE**: optional problems for students who want additional depth or a harder application.

The categories are repeated here so the worked answers map directly onto the practice sheet. Use the solutions to check method, units and physical interpretation rather than only the final number. A course-wide Symbols, Constants and Units sheet is provided separately; non-standard symbols are also defined locally.

Useful constants: Earth rotation rate $\Omega = 7.292 \times 10^{-5}$ s^{-1}. The Coriolis parameter is $f = 2\Omega \sin\phi$. Horizontal pressure-gradient acceleration may be estimated as $a_p = -(1/\rho)\,\Delta p/\Delta x$.

YOU SHOULD TRY Question 1: Identify the force balance from the situation

Complete the table by naming the dominant balance or additional force in each case. Then give one reason why the approximation might fail.

Situation	Dominant balance / force	One limitation
Vertical atmosphere outside a strong convective updraught		
Straight upper-level flow at 50°N, well above surface friction		
Wind at 10 m over rough land		
Flow very close to the equator		

Worked answer

For the first row, gravity is approximately balanced by the vertical pressure-gradient force: hydrostatic balance. It can fail locally where vertical acceleration is large, for example in strong convection.

For the second row, the horizontal pressure-gradient force can approximately balance Coriolis acceleration: geostrophic balance. Curvature, rapid evolution and residual friction can make the flow ageostrophic.

At 10 m, friction is an essential additional force. The wind is therefore slower than geostrophic and crosses isobars toward lower pressure.

Near the equator, f becomes small, so ordinary mid-latitude geostrophic balance is a poor approximation and other accelerations become relatively important.

YOU SHOULD TRY Question 2: Horizontal pressure-gradient acceleration

Sea-level pressure is 1014 hPa to the south of a region and 1008 hPa to the north, separated by 300 km. Take air density to be 1.20 $\mathrm{kg\,m^{-3}}$.

1. Calculate the magnitude of the horizontal pressure-gradient acceleration.
2. State its direction.
3. If the same 6 hPa pressure change occurred across only 150 km, by what factor would the acceleration change?

Worked answer

The pressure difference is 600 Pa and the distance is 3.00×10^5 m, so

$$|a_p| = \frac{600}{1.20(3.00 \times 10^5)} = 1.67 \times 10^{-3}\ \mathrm{m\,s^{-2}}.$$

The acceleration is from high toward low pressure, hence northward.

Halving the distance doubles $|\Delta p/\Delta x|$, so the acceleration would be twice as large.

YOU SHOULD TRY Question 3: Coriolis acceleration over Manchester

At latitude 53.5°N, calculate the Coriolis parameter f and the magnitude of the Coriolis acceleration for a wind speed of 15 $\mathrm{m\,s^{-1}}$.

1. What is the magnitude of f?
2. What is the acceleration magnitude?
3. If the wind is blowing due east, in which compass direction does the Coriolis acceleration point?

Worked answer

$$f = 2\Omega \sin\phi = 2(7.292 \times 10^{-5}) \sin 53.5° = 1.17 \times 10^{-4}\ \mathrm{s^{-1}}.$$

Hence

$$a_c = fv = (1.17 \times 10^{-4})(15) = 1.76 \times 10^{-3}\ \mathrm{m\,s^{-2}}.$$

In the Northern Hemisphere Coriolis acceleration acts to the right of the motion. For an eastward wind, "right" is southward.

IN CLASS Question 4: Geostrophic wind speed and direction

Use the pressure gradient from Question 2 and the latitude from Question 3. Assume straight isobars and negligible friction.

1. Estimate the geostrophic wind speed.
2. Determine whether the geostrophic wind is approximately eastward or westward.
3. State where the lower pressure lies relative to the geostrophic wind in the Northern Hemisphere.

Worked answer

Geostrophic balance requires

$$fv_g = \frac{1}{\rho}\frac{|\Delta p|}{\Delta x}.$$

Therefore

$$v_g = \frac{1.67 \times 10^{-3}}{1.17 \times 10^{-4}} = 14.2\ \mathrm{m\,s^{-1}}.$$

The pressure-gradient acceleration is northward, so Coriolis must act southward. In the Northern Hemisphere that requires an eastward geostrophic wind. Low pressure therefore lies to the left of the flow.

EXTRA CHALLENGE Question 5: Use isobar spacing to compare winds

Region A has isobars every 4 hPa separated by 80 km. Region B has the same 4 hPa interval separated by 200 km. Assume the same latitude and density.

1. Calculate the ratio of the pressure-gradient magnitudes, $|\nabla p|_A/|\nabla p|_B$.
2. Hence estimate the ratio of geostrophic wind speeds.
3. If the geostrophic wind in Region B is 12 $\mathrm{m\,s^{-1}}$, estimate it in Region A.
4. Explain why the actual 10-m wind in either region would usually be weaker and would cross the isobars toward lower pressure.

Worked answer

For a fixed pressure interval, pressure-gradient magnitude varies as inverse isobar spacing:

$$\frac{|\nabla p|_A}{|\nabla p|_B} = \frac{200}{80} = 2.5.$$

At the same f and density, geostrophic wind is proportional to the pressure gradient, so the wind-speed ratio is also 2.5. Thus

$$v_{g,A} = 2.5(12) = 30\ \mathrm{m\,s^{-1}}.$$

Near the surface, friction slows the wind. Since Coriolis acceleration is proportional to wind speed, it is reduced while the pressure-gradient acceleration remains, leaving a component of acceleration toward lower pressure. Surface winds therefore cross isobars and are usually weaker than the geostrophic value.

YOU SHOULD TRY Question 6: Diagnose a sea-breeze circulation from pressure data

During a sunny afternoon, measurements give the following idealised pressures:

	Over land	Over sea
Surface	1007 hPa	1010 hPa
At 1.5 km	850 hPa	846 hPa

1. At the surface, in which direction does the pressure-gradient acceleration act?
2. At 1.5 km, in which direction does it act?
3. Use these two answers to sketch the circulation in a vertical land-sea cross-section, including rising and sinking branches.
4. Why is this circulation normally strongest after several hours of daytime heating rather than just after sunrise?

Worked answer

At the surface, pressure is higher over the sea, so the pressure-gradient acceleration is from sea toward land: the sea-breeze inflow.

At 1.5 km, pressure is higher over the warmer land column, so the acceleration is from land toward sea: the return flow aloft.

Mass continuity closes the circulation with rising motion over the warmer land and sinking motion over the cooler sea. The circulation strengthens only after the land-sea temperature contrast has had time to build, so it is commonly best developed later in the day rather than immediately after sunrise.

YOU SHOULD TRY Question 7: From a surface pressure system to vertical motion and weather

Two idealised Northern Hemisphere pressure systems are observed.

System	Near-surface wind	Central pressure tendency
A	spirals inward anticlockwise	low pressure
B	spirals outward clockwise	high pressure

For each system, state whether the surface flow converges or diverges, infer the sign of the vertical motion required by mass continuity, and predict the broad cloud/precipitation tendency if moisture is available.

Worked answer

System A is a cyclone. Inward surface flow converges, so mass continuity requires net ascent above the convergence. Rising air expands and cools, increasing relative humidity and favouring cloud and precipitation when moisture is available.

System B is an anticyclone. Outward surface flow diverges, so compensating subsidence is expected. Descending air is compressed and warmed, lowering relative humidity and generally suppressing cloud and precipitation. Winter fog or low cloud can still occur beneath strong stable inversions.

EXTRA CHALLENGE Question 8: Use the three-cell circulation to diagnose latitude bands

For each latitude band in the table, fill in the most likely broad vertical motion and the prevailing surface-wind family in the Northern Hemisphere.

Latitude	Vertical motion	Prevailing surface wind
0°	?	?
30°N	?	?
50°N	?	?
60°N	?	?
90°N	?	?

Then identify which of these latitude bands is most strongly associated with (i) persistent tropical rainfall and (ii) subtropical desert belts.

Worked answer

Near the equator, broad ascent occurs in the ITCZ and near-surface trade winds converge from the subtropics. Near 30°N, the descending branch of the Hadley cell favours subtropical high pressure; surface flow is divided between equatorward trade winds and poleward flow feeding the mid-latitude westerlies. Around 50°N, prevailing surface flow is broadly westerly. Near 60°N, ascent is favoured near the subpolar low/polar-front region where westerlies meet polar easterlies. Near the pole, cold air broadly subsides and surface polar easterlies flow equatorward.

The equatorial ascending belt is associated with persistent tropical rainfall. The descending subtropical belt near 20–35° is associated with many major deserts.

EXTRA CHALLENGE Question 9: Temperature contrast and the jet stream

Two winter air columns have the same surface pressure. At a fixed pressure level aloft, the surface-to-level thickness is 5.7 km in the warm column but 5.2 km in the cold column.

1. Which column places that pressure surface at the greater geometric height?
2. What does the 0.5 km height difference imply about the horizontal pressure-gradient tendency aloft across the boundary between the columns?
3. Use geostrophic reasoning to explain why a strong upper-level wind tends to occur near a strong horizontal temperature contrast, and why the mid-latitude jet is often stronger in winter.

Worked answer

The pressure surface is higher in the warm column because warmer air has a greater thickness between pressure levels. The resulting slope of pressure surfaces means that the horizontal pressure gradient becomes stronger with height across the warm-cold boundary. Geostrophic balance converts this upper-level pressure gradient into strong flow approximately parallel to the pressure contours. Winter equator-to-pole temperature contrasts are generally larger, so the associated upper-level pressure gradients and jet-stream winds can also be stronger.

Modelling connection.
These harder dynamics problems show how geostrophic balance, curved flow and Rossby-wave ideas appear in simple atmospheric models. Students who also take EART22001 will meet related ideas in a shallow-water model, but no knowledge of that unit is assumed here.

EXTRA CHALLENGE Question 10: Gradient wind around a curved low-pressure system

At latitude 50°N, an air parcel follows circular isobars around a low with radius of curvature $R = 500$ km. The inward pressure-gradient acceleration is $a_p = 2.0 \times 10^{-3}\ \mathrm{m\,s^{-2}}$. Neglect friction. For cyclonic flow in the Northern Hemisphere use the radial balance

$$\frac{v^2}{R} + fv = a_p, \qquad f = 2\Omega \sin\phi.$$

1. Calculate f.
2. Solve the quadratic for the physically relevant gradient-wind speed v.
3. Calculate the geostrophic speed $v_g = a_p/f$ for the same pressure gradient.
4. Is the cyclonic gradient wind subgeostrophic or supergeostrophic here? Explain using the force balance.

Worked answer

At 50°N,

$$f = 2(7.292 \times 10^{-5}) \sin 50^\circ = 1.12 \times 10^{-4}\ \mathrm{s^{-1}}.$$

The balance gives

$$\frac{v^2}{5.0 \times 10^5} + (1.12 \times 10^{-4})v - 2.0 \times 10^{-3} = 0.$$

Multiplying by R,

$$v^2 + 56.0v - 1000 = 0,$$

so the positive root is

$$v = \frac{-56.0 + \sqrt{56.0^2 + 4000}}{2} = 14.5\ \mathrm{m\,s^{-1}}.$$

The geostrophic speed is

$$v_g = \frac{2.0 \times 10^{-3}}{1.12 \times 10^{-4}} = 17.9 \text{ m s}^{-1}.$$

Thus the cyclonic gradient wind is subgeostrophic. Part of the inward pressure-gradient force supplies the centripetal acceleration, so a smaller Coriolis force - and hence a smaller wind speed - is needed than in straight geostrophic flow.

EXTRA CHALLENGE Question 11: Thermal wind: why jets strengthen with horizontal temperature contrast

At 50°N, the mean temperature at 850 hPa decreases northward by 8 K over 1000 km. Use the pressure-coordinate thermal-wind estimate

$$\Delta u_g \approx -\frac{R_d}{f}\frac{\partial T}{\partial y}\ln\left(\frac{p_2}{p_1}\right),$$

with $p_1 = 850$ hPa, $p_2 = 300$ hPa and $f = 1.12 \times 10^{-4}$ s^{-1}.

1. Estimate $\partial T/\partial y$ in K m^{-1}.
2. Estimate the change in geostrophic zonal wind between 850 and 300 hPa.
3. Explain why a stronger equator-to-pole temperature gradient in winter is associated with a stronger mid-latitude jet.

Worked answer

Northward temperature decreases, so

$$\frac{\partial T}{\partial y} = \frac{-8}{1.0 \times 10^{6}} = -8.0 \times 10^{-6} \text{ K m}^{-1}.$$

Also,

$$\ln\left(\frac{300}{850}\right) = -1.042.$$

Hence

$$\Delta u_g \approx -\frac{287}{1.12 \times 10^{-4}}(-8.0 \times 10^{-6})(-1.042) \approx -21.4 \text{ m s}^{-1}.$$

The sign depends on the coordinate convention; the key magnitude is about 21 m s^{-1} of vertical geostrophic shear across this layer. A larger horizontal temperature gradient gives a larger thermal-wind shear, helping produce stronger upper-level westerly jets in winter.

IN CLASS Question 12: Orographic Rossby-wave reasoning

A barotropic westerly flow crosses an isolated north-south mountain ridge. The flow is displaced poleward on one side of the disturbance and equatorward on the other. Assume the Coriolis parameter increases with latitude.

1. If a parcel moves poleward while conserving absolute vorticity approximately, how must its relative vorticity change?
2. What happens for an equatorward displacement?
3. Explain qualitatively how alternating meridional displacements can support a Rossby-wave pattern downstream of topography.
4. Why is geostrophic balance a useful starting point for initialising such a shallow-water experiment?

Worked answer

Absolute vorticity is approximately $\zeta + f$. A poleward displacement increases f, so relative vorticity ζ must become more negative (more anticyclonic) if the sum is approximately conserved. An equatorward displacement decreases f, so ζ becomes more positive (more cyclonic).

These restoring changes in relative vorticity cause the flow to curve back and can produce alternating ridges and troughs downstream: a Rossby-wave response. Starting from geostrophic balance minimises spurious fast adjustment; the initial pressure/height field and velocity field are already close to dynamical balance before the orographic perturbation acts.